UNIT - IV

DEFLECTION OF BEAMS & TORSION

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- DEFLECTION: bending into a circular arc, slope, deflection, radius of curvature
- Differential equation of the elastic line, double integration and Macaulay's methods
- Slope and deflection of cantilever and simply supported beams: point load, u.d.l., u.v.l.
- Mohr's theorems, moment-area method
- TORSION: derivation, torsion of circular shafts, pure shear
- Power transmission by circular shafts, shafts in series and parallel

Tip: memorise the standard slope and deflection table - it saves a lot of time in exams.

PART A - DEFLECTION OF BEAMS

1. Introduction

Under load a beam bends. The deflected shape of its neutral axis is the elastic curve. Limits on deflection are needed for serviceability (cracking of plaster, machine alignment). A common limit is $\delta_{max} \le L / 325$ for floors.

- Deflection y: vertical displacement of a point on the N.A. from its original position.

- Slope θ: angle between the tangent to the elastic curve and the original axis; $\theta \approx dy/dx$ for small deflections.
- Flexural rigidity EI: resistance of the beam to bending.

2. Bending into a circular arc

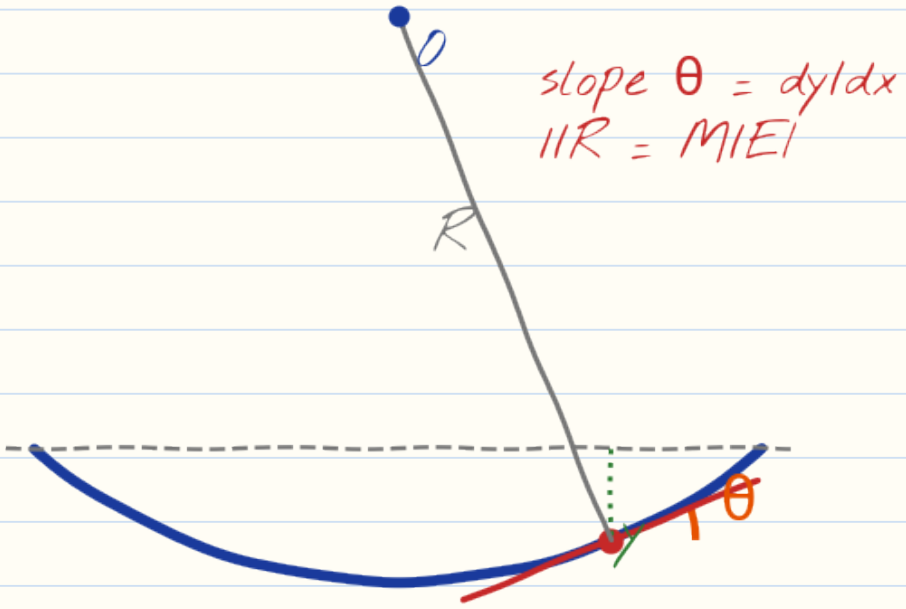


Fig: elastic curve, slope and radius of curvature R

From simple bending $M/I = E/R$, so the curvature is:

$$1/R = M/EI$$

From geometry the exact curvature of the curve $y(x)$ is $1/R = y''/[1 + (y')^2]^{3/2}$. For small slopes $(y')^2$ is negligible, so $1/R \approx d^2y/dx^2$.

3. Differential equation of the elastic line

$$EI\, d^2y/dx^2 = M(x)$$

Sign: with y measured upward positive and x to the right, a sagging (+ve) moment gives a curve concave upward, $y'' > 0$. Downward deflection therefore comes out negative - we quote the magnitude.

- $EI\, dy/dx = \int M\, dx + C_1$ (slope equation)

- $EIy = \iint M\,dx\,dx + C_1 x + C_2$ (deflection equation)
- C_1, C_2 from boundary conditions.

Support	Boundary condition
Fixed end	$y = 0$ and $\theta = 0$
Simple support (pin or roller)	$y = 0$
Free end	$M = 0$, $V = 0$
Symmetric beam, centre	$\theta = 0$

4. Double integration method

- 1. Find reactions and write $M(x)$ for the section at distance x (one expression per portion).
- 2. Put $M(x)$ in $EIy'' = M$ and integrate twice.
- 3. Apply boundary / continuity conditions to get C_1, C_2.
- 4. Substitute x to get slope or deflection; maximum deflection is where slope = 0.

Example 1: cantilever, point load W at free end

Take x from the fixed end: $M = -W(L - x)$.

$EIy'' = -W(L - x)$

$EIy' = -W(Lx - x^2/2) + C_1$; $y' = 0$ at $x = 0 \Rightarrow C_1 = 0$

$EIy = -W(Lx^2/2 - x^3/6) + C_2$; $y = 0$ at $x = 0 \Rightarrow C_2 = 0$

At $x = L$: $\theta_B = -WL^2/2EI$, $y_B = -WL^3/3EI$ (downward)

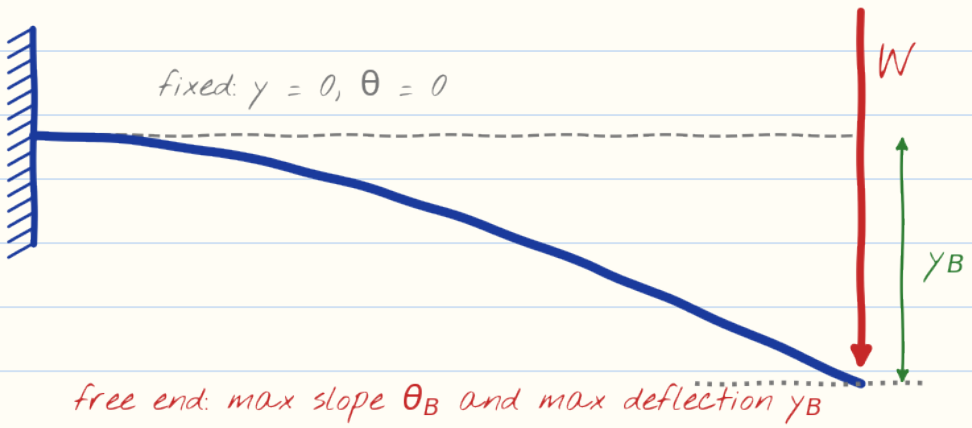


Fig: cantilever - maximum slope and deflection at the free end

Example 2: simply supported beam with u.d.l.

$M = w L x/2 - w x^2/2$

$E I y' = w L x^2/4 - w x^3/6 + C_1$

$E I y = w L x^3/12 - w x^4/24 + C_1 x + C_2$

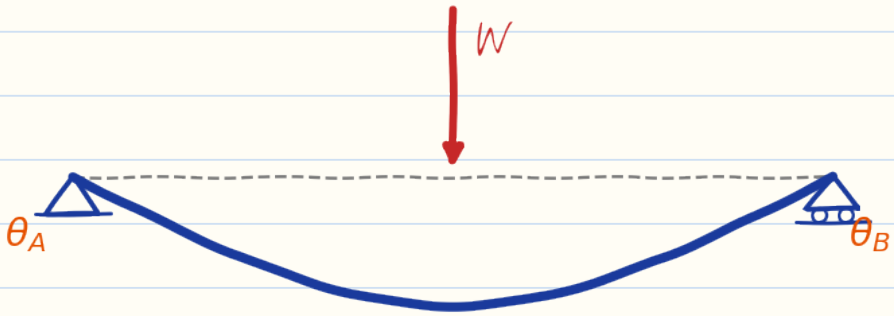
$y = 0$ at $x = 0 \Rightarrow C_2 = 0$; $y = 0$ at $x = L \Rightarrow C_1 = - w L^3/24$

Slope at A ($x = 0$): $\theta_A = - w L^3 / 24EI$

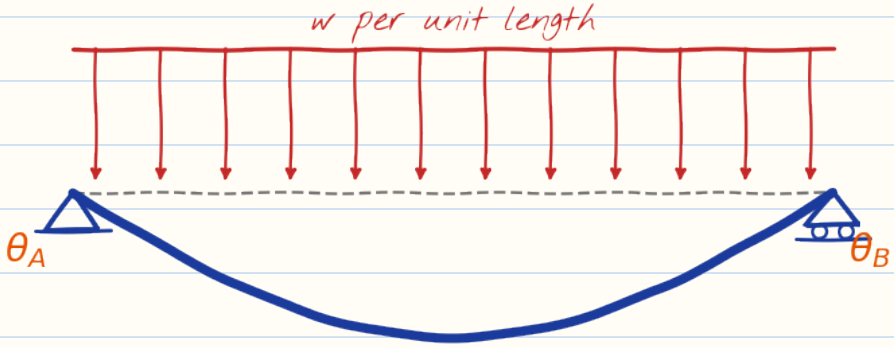
At centre $x = L/2$: $E I y = wL^4 [1/96 - 1/384 - 1/48] = - 5 w L^4/384$

$$y_{max} = 5 w L^4 / 384 E I$$

5. Standard results



$y_{max} = WL^3 / 48EI$ $\theta_A = \theta_B = WL^2 / 16EI$



$y_{max} = 5wL^4 / 384EI$ $\theta_A = \theta_B = wL^3 / 24EI$

Fig: S.S. beam - central point load and u.d.l.

Beam and loading	Slope (max)	Deflection (max)
Cantilever, W at free end	$WL^2 / 2EI$	$WL^3 / 3EI$
Cantilever, u.d.l. w	$wL^3 / 6EI$	$wL^4 / 8EI$
Cantilever, u.v.l. (0 at free end)	$wL^3 / 24EI$	$wL^4 / 30EI$
Cantilever, couple M at free end	ML / EI	$ML^2 / 2EI$
S.S., central W	$WL^2 / 16EI$	$WL^3 / 48EI$
S.S., u.d.l. w	$wL^3 / 24EI$	$5wL^4 / 384EI$
S.S., W at a from A, b from B	$\theta_A = Wb(L^2-b^2) / 6LEI$	under load: $Wa^2b^2 / 3EIL$

For a S.S. beam with u.v.l. (0 at A, w_0 at B): $\theta_A = 7w_0L^3/360EI$, $\theta_B = w_0L^3/45EI$, $y_{max} = 0.00652\, w_0L^4/EI$ at $x = 0.519L$.

6. Macaulay's method

Double integration needs a new pair of constants for every portion of a beam with several loads. Macaulay's method writes a single expression for M valid over the whole beam, using brackets $[x - a]$ that vanish when $x < a$. Then only two constants appear.

- Take the origin at the left end.
- Write M from the left, always on the section at the last portion (x in the right-most part).
- Write a term for each load: $W[x - a]$, $w[x - a]^2/2$, couple $M_0[x - a]^0$.
- A u.d.l. that stops before the end is extended to the end and an equal opposite u.d.l. is added.
- Integrate the brackets as a whole: $\int [x - a]\, dx = [x - a]^2/2$ (do NOT expand).
- A bracket is ignored if its value is negative ($x < a$).

Example 3: S.S. beam, point load W at a from A (b = L - a)

$R_A = W b / L$

$E I y'' = R_A x - W [x - a]$

$E I y' = R_A x^2/2 - W [x - a]^2/2 + C_1$

$E I y = R_A x^3/6 - W [x - a]^3/6 + C_1 x + C_2$

$x = 0, y = 0 \Rightarrow C_2 = 0.$ $x = L, y = 0 \Rightarrow C_1 = - W b (L^2 - b^2) / 6L$

So $\theta_A = - C_1/EI = W b (L^2 - b^2) / 6 L E I$ and under the load $y = W a^2 b^2 / 3 E I L$

Example 4: numerical

S.S. beam, span 5 m, point load 10 KN at 2 m from A. $E I = 1 \times 10^4$ KN·m². Find deflection under the load and slope at A.

$a = 2, b = 3, L = 5.$ $y = W a^2 b^2 / 3 E I L = 10 \times 4 \times 9 / (3 \times 1 \times 10^4 \times 5) = 2.4 \times 10^{-3}$ m = 2.4 mm

$\theta_A = W b (L^2 - b^2) / 6 L E I = 10 \times 3 \times 16 / (6 \times 5 \times 1 \times 10^4) = 1.6 \times 10^{-3}$ rad

7. Moment-area method (Mohr's theorems)

Draw the M/EI diagram. Between two points A and B of the elastic curve:

- Mohr's theorem I: the change of slope between A and B = area of the M/EI diagram between A and B. $\theta_{AB} = A_{AB}$
- Mohr's theorem II: the deviation of B from the tangent at A (measured perpendicular to the beam) = moment of the M/EI area between A and B about B. $\Delta_B = A_{AB} \cdot \bar{x}_B$

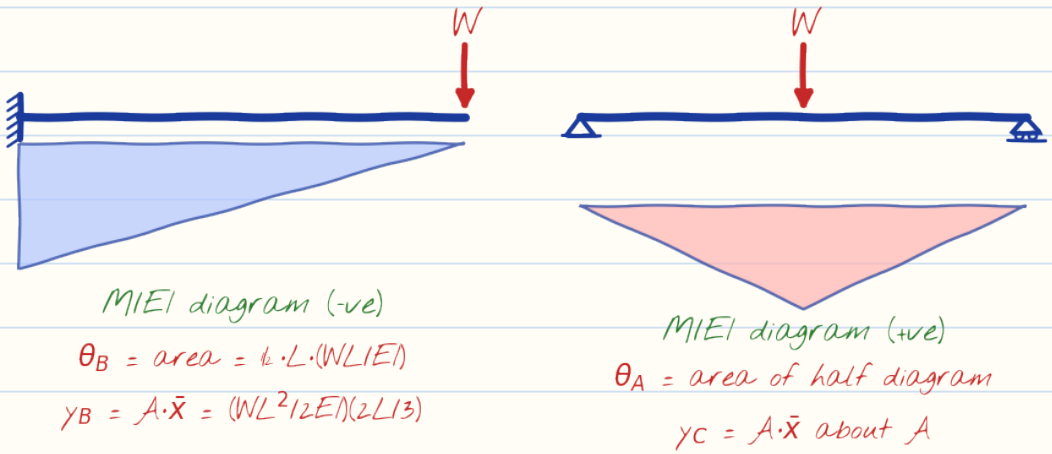


Fig: moment-area applications

Example 5: cantilever, W at free end

M/EI is a triangle: height W L / EI at the fixed end, area = ½ · L · W L/EI = W L²/2EI = θ_B

Centroid is 2L/3 from the free end: y_B = (W L²/2EI)(2L/3) = W L³/3EI

Example 6: S.S. beam, central W

Half of the M/EI diagram: area = ½ (L/2)(W L/4EI) = W L²/16EI = θ_A (tangent at C is horizontal)

y_C = moment of this area about A = (W L²/16EI)(L/3) = W L³/48EI

PART B - TORSION

8. Introduction

Torsion is the twisting of a member under a torque (twisting moment) T whose vector is along the axis - e.g. shafts of motors, turbines, drive shafts.

- Assumptions: the shaft is straight, circular and of uniform section; material homogeneous and elastic;
- plane cross-sections stay plane and radii stay straight after twisting;

- twist is uniform along the length, shear stress is proportional to shear strain ($\tau = G\gamma$).

9. Derivation of the torsion equation

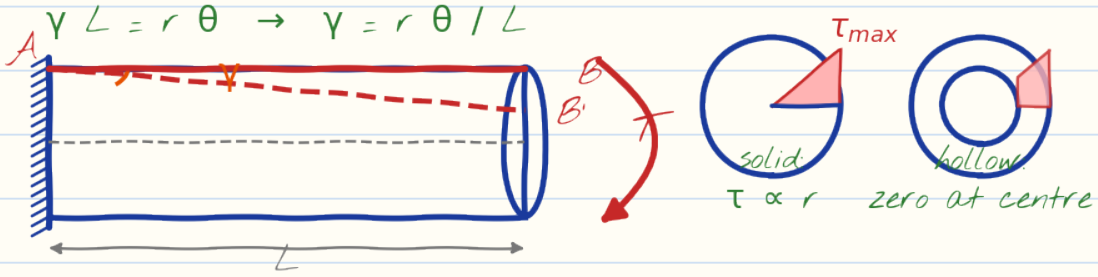


Fig: twisting of a shaft, shear stress on solid and hollow sections

A line AB on the surface moves to AB' – shear strain γ. For radius r and angle of twist θ over length L:

$\gamma L = r\theta \Rightarrow \gamma = r\theta / L$

$\tau = G\gamma = G r\theta / L \Rightarrow \tau / r = G\theta / L$ ...(1)

Moment of the force $\tau\, dA$ about the axis: $T = \int \tau r\, dA = (G\theta / L)\int r^2\, dA = G\theta J / L$ ...(2)

$$T / J = \tau / r = G\theta / L$$

- J = polar M.I. of the section (mm^4), G = modulus of rigidity, θ in radians.
- Solid shaft: $J = \pi d^4/32$; hollow: $J = \pi (D^4 - d^4)/32$.
- Torsional rigidity = GJ; polar modulus $Z_p = J / R$ (maximum shear stress $\tau_{max} = T / Z_p$).

Solid: $T = \pi \tau d^3 / 16$ Hollow: $T = \pi \tau (D^4 - d^4) / 16D$

10. Pure shear

A shaft element is in a state of **pure shear**: shear stress τ on the faces with no normal stress. Using the Mohr's circle (centre at the origin, radius τ): the principal stresses are $\sigma_1 = +\tau$ and $\sigma_2 = -\tau$ on planes at 45°.

- Ductile materials (mild steel) fail in shear on a plane perpendicular to the axis.
- Brittle materials (cast iron, chalk) fail along a 45° helix - on the plane of maximum tension.
- The relation $E = 2G(1+\mu)$ follows from this state.

11. Power transmitted by a shaft

$$P = T\omega = 2\pi N T / 60 \text{ (watts)} \qquad T = 60 P / (2\pi N)$$

N in rpm, T in N·m, P in W. (In KW: $T = 9550\, P / N$ N·m.)

Example 7

A solid shaft transmits 100 KW at 200 rpm. $\tau_{allow} = 60$ MPa, $G = 80$ GPa, $L = 2$ m. Find d and the angle of twist.

$T = 100\times10^3 \times 60 / (2\pi \times 200) = 4774.6$ N·m $= 4.7746\times10^6$ N·mm

$d^3 = 16T / (\pi\tau) = 16 \times 4.7746\times10^6 / (\pi \times 60) = 4.053\times10^5 \Rightarrow d = 74$ mm $\Rightarrow$ use 75 mm

$J = \pi \times 75^4 / 32 = 3.106\times10^6$ mm^4

$\theta = TL / GJ = 4.7746\times10^6 \times 2000 / (8\times10^4 \times 3.106\times10^6) = 0.0384$ rad $= 2.2°$

Example 8: hollow shaft

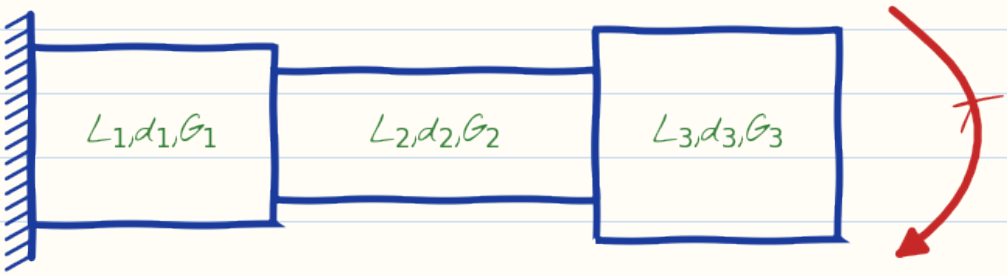
A hollow shaft $D = 100$ mm, $d = 80$ mm transmits T. If $\tau_{allow} = 50$ MPa find T.

$J = \pi(100^4 - 80^4)/32 = 5.797\times10^6$ mm^4, $Z_p = J/50 = 1.159\times10^5$ mm^3

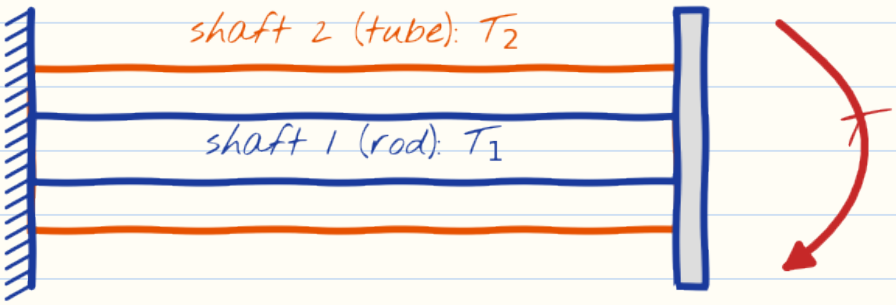
$T = \tau Z_p = 50 \times 1.159\times10^5 = 5.797\times10^6$ N·mm $= 5.80$ KN·m

A hollow shaft is more economical than a solid one: material near the axis carries little stress.

12. Shafts in series and in parallel



SERIES: same T in each, $\theta = \Sigma\, T L / G J$



PARALLEL: same θ, $T = T_1 + T_2$

- Series: shafts of different sections joined end to end carry the same torque. Total twist $\theta = \Sigma\, T L / (G J) = T L_1/G_1J_1 + T L_2/G_2J_2 + \dots$
- Parallel: shafts joined at both ends (e.g. a tube over a rod) have the same angle of twist, the torque is shared: $T = T_1 + T_2$.

$$T_1 / T_2 = (G_1J_1 / L_1) / (G_2J_2 / L_2) \qquad T_1 = T\, G_1J_1 / (G_1J_1 + G_2J_2) \text{ (same } L)$$

Example 9: shafts in parallel

A steel rod ($J_1 = 2\times10^6$ mm^4, $G = 80$ GPa) is inside a bronze tube ($J_2 = 6\times10^6$ mm^4, $G = 40$ GPa), same length, joined at ends. $T = 4$ KN·m. Find torque shares.

$G_1J_1 = 1.6\times10^{11}$, $G_2J_2 = 2.4\times10^{11}$

$T_1 = 4 \times 1.6/4.0 = 1.6$ KN·m, $T_2 = 2.4$ KN·m

More solved problems

Example 10: cantilever with u.d.l.

Cantilever $L = 2$ m, u.d.l. 10 KN/m, $E = 200$ GPa, $I = 8\times10^7$ mm^4. Find free-end slope and deflection.

$w = 10$ N/mm, $L = 2000$ mm, $EI = 2\times10^5 \times 8\times10^7 = 1.6\times10^{13}$ N·mm^2

$y_B = wL^4 / 8EI = 10 \times 1.6\times10^{13} / (8 \times 1.6\times10^{13}) = 1.25$ mm

$\theta_B = wL^3 / 6EI = 10 \times 8\times10^9 / (6 \times 1.6\times10^{13}) = 8.33\times10^{-4}$ rad

Example 11: S.S. beam, check against L/325

S.S. beam span 6 m, u.d.l. 12 KN/m, $E = 200$ GPa, $I = 1.5\times10^8$ mm^4.

$y_{max} = 5wL^4 / 384EI = 5 \times 12 \times (6000)^4 / (384 \times 2\times10^5 \times 1.5\times10^8) = 6.75$ mm

Permissible $= L/325 = 18.5$ mm $\Rightarrow$ safe in deflection ✓

Example 12: shaft - stress and twist

Solid shaft $d = 50$ mm, $L = 1.5$ m, $T = 1$ KN·m, $G = 80$ GPa.

$\tau = 16T / \pi d^3 = 16\times10^6 / (\pi \times 125\,000) = 40.7$ MPa

$J = \pi \times 50^4 / 32 = 6.136\times10^5$ mm^4

$\theta = TL / GJ = 10^6 \times 1500 / (8\times10^4 \times 6.136\times10^5) = 0.0306$ rad $= 1.75°$

Example 13: stepped shaft (series)

Shaft AB: $d = 60$ mm, $L = 1$ m; BC: $d = 40$ mm, $L = 1$ m; $T = 500$ N·m; $G = 80$ GPa.

$J_1 = 1.272\times10^6$, $J_2 = 2.513\times10^5$ mm^4

$\theta = (T L / G)(1/J_1 + 1/J_2) = 6.25 \times 4.765 \times 10^{-6} = 0.0298$ rad = 1.71°

Maximum shear stress is in the thinner part BC: $\tau = 16 \times 5 \times 10^5 / (\pi \times 40^3) = 39.8$ MPa

Strength and stiffness criteria

- A shaft is designed for strength ($\tau \leq \tau_{allow}$) and checked for stiffness (θ per metre $\leq$ limit, often 0.25° to 1° per metre).
- Take the larger diameter obtained from the two criteria.
- Beams: design for bending (strength), then verify the deflection limit L/325 or as specified.
- Deflection varies with L^3 or L^4 - doubling the span makes the beam far more flexible; doubling depth reduces deflection to 1/8.

Important questions

- Derive the differential equation of the elastic curve: $E I \, d^2y/dx^2 = M$.
- Find slope and deflection of a cantilever with u.d.l. using double integration.
- Explain Macaulay's method; find the deflection under the load of a S.S. beam.
- State and prove Mohr's theorems; use them to find the deflection of a cantilever.
- Derive $T/J = \tau/r = G\theta/L$ with assumptions.
- Compare solid and hollow shafts. A shaft transmits 150 KW at 300 rpm - find the diameter.

Quick revision - Unit IV

$EI\,y'' = M$ $\quad 1/R = M/EI$ $\quad \theta = \int M/EI\,dx$ $\quad y_{max}$ (S.S., udl) $= 5wL^4/384EI$

$T/J = \tau/r = G\theta/L$ $\quad P = 2\pi NT/60$ $\quad$ series: same T $\quad$ parallel: same θ

- Important: derive $y = WL^3/3EI$ (cantilever) and $5wL^4/384EI$ (S.S. udl) by double integration.
- Important: Macaulay's method problems; moment-area theorems; torsion equation derivation; power transmission; shafts in series/parallel.